

General Features

The i012 and i014 are DC/DC converters for the ionia™ system. They supply the central units (e.g. i101, i102, i103) and all additional communication and I/O modules via the backplane.

While i012 can supply up to 20W, the i014 delivers up to 40 W. Any ionia™ system configuration is supported by one of these two DC/DC converters.

These modules are connected directly to the train battery and support all voltage types from 24 to 110Vdc. Voltage variations down to 14.4Vdc and up to 154Vdc are fully supported.

A built-in supervisory mechanism called «Condition Monitoring» can detect, record and report any off-limit usage, such as overvoltage, overcurrent, supply interruption, overtemperature, etc. as well as selected user-defined values.

The modules offer a fast and secure point-to-point connection to the central unit, which allows an instant propagation of all events of interest.

Monitor and recorder configuring is done easily and time saving by one of the five methods defined by IEC 61131.

i012 and i014 are designed for harsh rolling stock environments and are fully compliant to EN 50155.

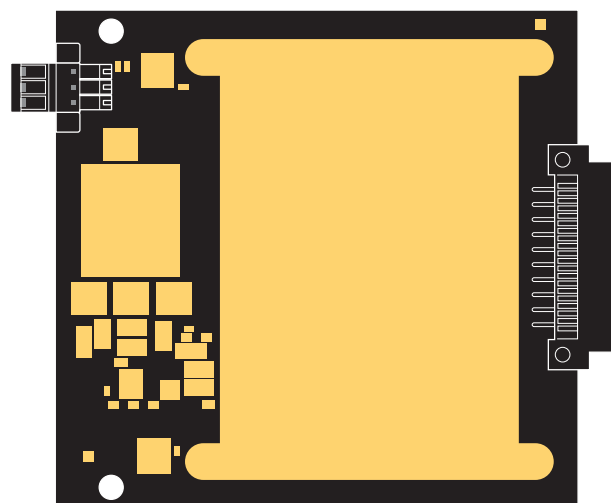
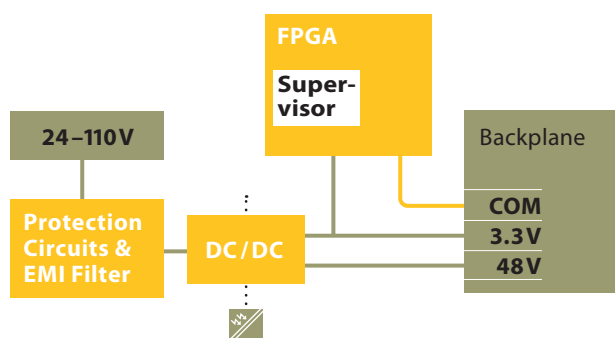
Key Benefits

- One single device, no need for additional external DC/DC converters
- No capacitor replacement necessary
- «Condition Monitoring» gives detailed diagnostics to the central unit
- IEC 61131 for fast and easy monitoring and recording configuration
- Complies to EN 50155, EN 50121, IEC 61373

Application Examples

- Any input/output/communication system
- PoE Switch without an additional external DC/DC converter
- WTB Gateway including fritting option without any additional DC/DC converter
- All Kind of communication gateways

i012 Hardware Architecture



Life Cycle Cost

The i012 and i014 have been strictly designed without electrolytic capacitor to avoid service costs for changing capacitors. An active overvoltage sensing circuit protects the device

against damage through excessive overvoltage. Moreover, the internal overcurrent protection enhances the life cycle of these power modules.

Technical Data

Input Voltage	– 24 Vdc to 110 Vdc – Reverse polarity, overvoltage and overtemperature safe – 1000 Vac galvanic isolation between train battery and the secondary side
Output Voltage / Output Power	– Delivering 48 Vdc to all communication and I/O modules – Sourcing the 3.3 V backplane – Active overcurrent protection – i012 $P_{max} = 20 W$ – i014 $P_{max} = 40 W$ – Interruption class S2 with ceramic capacitors
Configuration	– Programmable by the central unit using IEC 61131 compliant executor and IDE supporting LD, FBD, ST, IL and SFC
Security / Watchdog	– Communication Link with watchdog supervision – External supervisor for voltage, current and temp monitoring → «Condition Monitoring»
Backplane Communication	– 100 Mbit duagon serial star slave → no need for physical addressing – Secure communication protocol between modules and the central unit

Diagnostic/Service	– LEDs for visual status information (power on, running, communicating, failure ...) – Diagnostic data accessible through the central unit's service interfaces (see i01, i02 or i03 for more information about these possibilities)
Power Consumption	– P_{max} control: <1 W – Efficiency: 80% – Supports sleep mode
Operating Conditions	– Ambient temperature: – 40 °C to +85 °C (EN 50155, Class TX) – Relative humidity: – 75%, max 95% for 30 days per year (conformal coating) EN 60068 – Shock and vibration: – According to IEC 61373 category 1, class B – EMI: – According to EN 50121 and EN 50155
Dimensions/Weight	– 4TE x 92 x 86 mm, fitting the ionia™ housing – 150 g
Environment	– Fully compliant with RoHS and REACH

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Order Code and HW Options

Product Type:	i	-		EXAMPLE: i014-LV or i012-HV
Power			Battery Voltage	
20 W	i012		24 Vdc	LV
			36–48 Vdc	MV*
40 W	i014		72 Vdc	EV
			96–110 Vdc	HV

*Contact duagon for lead times and availability

Related Documents

Data Sheet i012/i014  d-006464-nnnnnn.pdf
available at www.duagon.com

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